

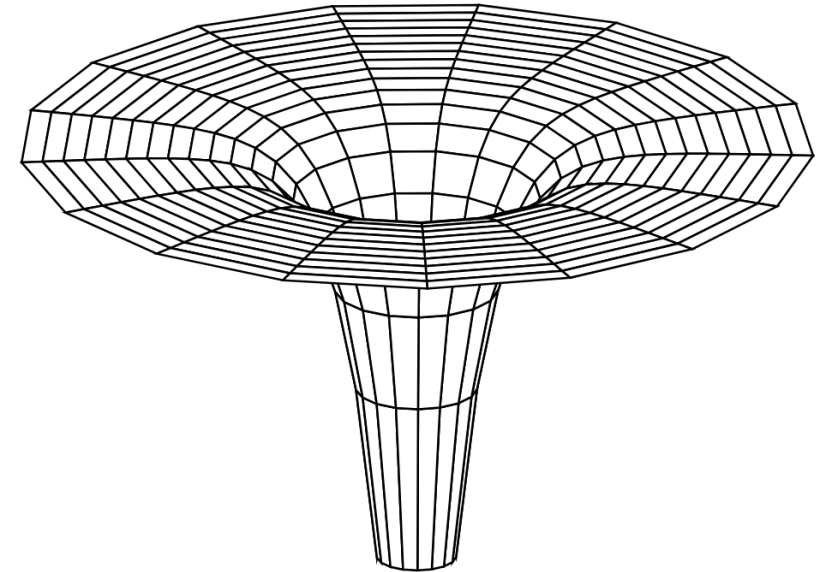
# Regularity of the Lorentzian Distance and Length Function at the Singular Boundary in the Kruskal Spacetime

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## What happens at the singularity?

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

- Singularities at  $r = r_s$  and  $r = 0$
- $M_{\text{int}} := (0, r_s) \times \mathbb{R} \times S^2 \Rightarrow \partial_t$  is spacelike
- $M_{\text{ext}} := (r_s, \infty) \times \mathbb{R} \times S^2 \Rightarrow \partial_t$  is timelike



## Kruskal Coordinates

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- For  $r \geq r_s$ :

$$T := \left(-1 + \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \sinh\left(\frac{t}{2r_s}\right), \quad X := \left(-1 + \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \cosh\left(\frac{t}{2r_s}\right)$$

- For  $r \leq r_s$ :

$$T := \left(1 - \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \cosh\left(\frac{t}{2r_s}\right), \quad X := \left(1 - \frac{r}{r_s}\right)^{\frac{1}{2}} e^{\frac{r}{2r_s}} \sinh\left(\frac{t}{2r_s}\right)$$

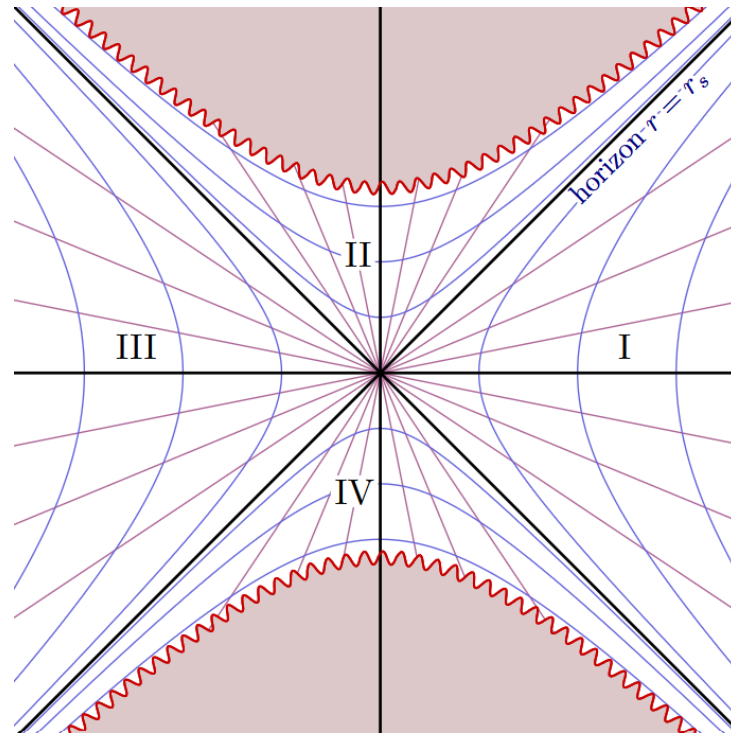
- Metric:  $ds^2 = \frac{4r_s^3}{r} e^{-\frac{r}{r_s}} (-dT^2 + dX^2) + r^2(d\theta^2 + \sin^2 \theta d\varphi^2)$

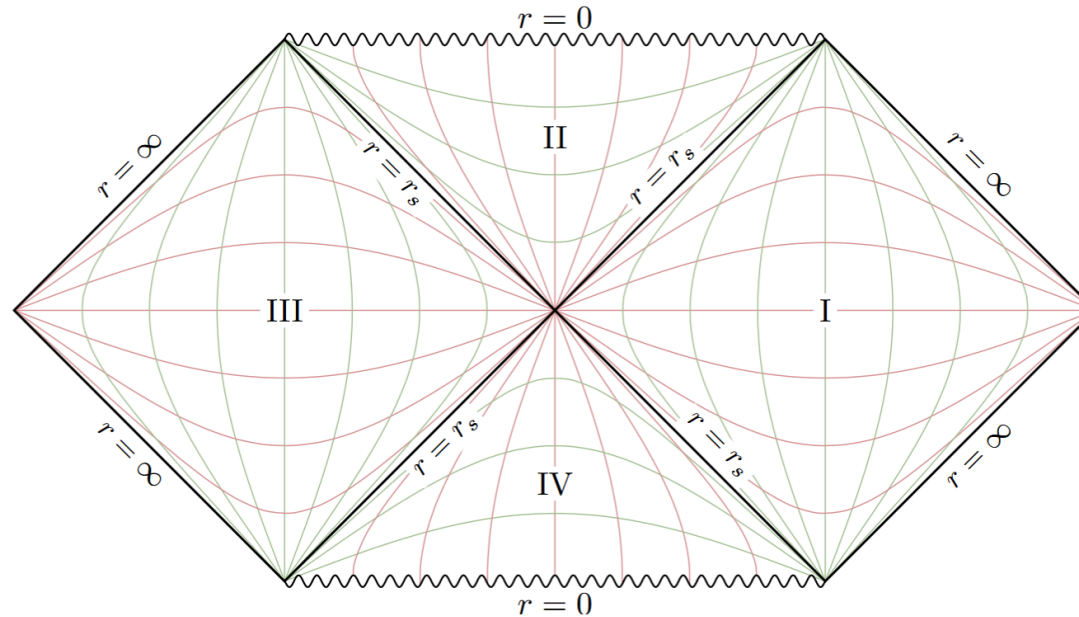
**Theorem**

Kruskal spacetime is isometrically inextendable.

*Proof sketch.*

Isometric inextendability follows from the divergence of tidal forces along all finite causal geodesics.





### Definition (Lorentzian Length and Distance)

$$\ell(\gamma) = \lambda(p, q) := \int_p^q \sqrt{-g(\dot{\gamma}(t), \dot{\gamma}(t))} dt$$

$$\tau(p, q) := \sup\{\ell(\gamma) \mid \gamma \text{ is a future-directed causal curve from } p \text{ to } q\}$$

## Geometric & Causal Foundation

### **Definition** (Global Hyperbolicity)

A time-oriented Lorentzian manifold  $(M, g)$  is called **globally hyperbolic** (g.h.) if it satisfies the following two conditions:

1.  $(M, g)$  is causal;
2.  $\forall p, q \in M$ , the set  $J^+(p) \cap J^-(q)$  is compact.

**Causal ladder:** chronological  $\rightarrow$  causal  $\rightarrow$  strongly causal  $\rightarrow$  *globally hyperbolic*

- $M_{\text{ext}}$  and  $M_{\text{int}}$  are globally hyperbolic when considered separately
- In a g.h. spacetime, for  $p \ll q$  there exists a maximizing geodesic realising  $\tau(p, q)$
- Killing vectors  $\partial_t$  and  $\partial_\varphi \rightarrow$  conserved quantities  $E$  and  $L$

$\rightarrow$  Restriction to future-directed timelike geodesics in  $M_{\text{int}}$ , with  $r$  arbitrarily close to 0

## Reduction to 1D integrals

$$E^2 = \left( \frac{dr}{d\lambda} \right)^2 + \mathcal{K}(r) \left( 1 + \frac{L^2}{r^2} \right), \quad \mathcal{K}(r) = 1 - \frac{r_s}{r}$$

- Reduction to  $\theta = \frac{\pi}{2}$  (by  $SO(3)$ -symmetry)
- On  $M_{\text{int}}$ ,  $r$  is timelike, so  $\frac{dr}{d\lambda} < 0$  along every future-directed timelike curve

→  $r$  strictly monotonic and can be used as a curve parameter → 1D Integral<sup>1</sup>

$$\begin{array}{c|c} L = r^2 \frac{d\varphi}{d\lambda} & E = \mathcal{K}(r) \frac{dt}{d\lambda} \\ \varphi(p, q) = \int_{r_q}^{r_p} \frac{L}{r^2} \frac{dr}{\sqrt{E^2 - \left(1 - \frac{r_s}{r}\right) \left(1 + \frac{L^2}{r^2}\right)}} & t(p, q) = \int_{r_q}^{r_p} \frac{E}{1 - \frac{r_s}{r}} \frac{dr}{\sqrt{E^2 - \left(1 - \frac{r_s}{r}\right) \left(1 + \frac{L^2}{r^2}\right)}} \end{array}$$

<sup>1</sup>Barrett O'Neill, *Semi-Riemannian Geometry : with Applications to Relativity*, 1st ed. (Academic Press, 1983).

## The Future Completion

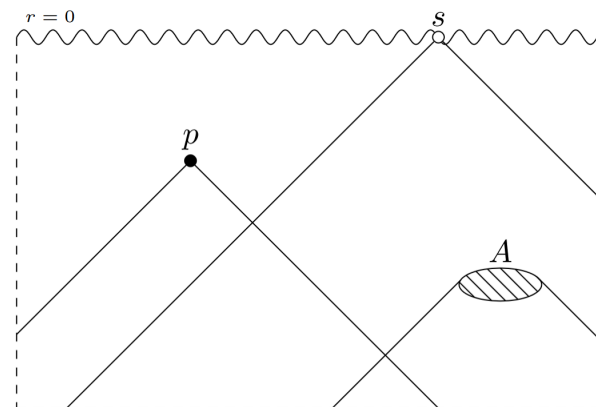
### Theorem

The interior  $M_{\text{int}} = (r_s, 0) \times \mathbb{R} \times S^2$  admits a future completion that is chronological homeomorphic to  $\overline{M}_{\text{int}} = (r_s, 0] \times \mathbb{R} \times S^2$  which includes ideal boundary points.

*Proof.* The proof can be found in

*Topologies on the future causal completion<sup>2</sup>*

by Dr. Olaf Müller.



□

<sup>2</sup>Olaf T. Müller, “Topologies on the Future Causal Completion,” 2024, <https://arxiv.org/abs/1909.03797>.



## Model Integral

### Proposition

Let  $P(x) = ax + bx^2 + cx^3$  with  $c \neq 0$ . On every interval on which  $|P(x)| < 1$ , the integral

$$I_\alpha(x) = \int dx x^\alpha (1 + P(x))^{-\frac{1}{2}}$$

admits a convergent power series representation of the form

$$I_\alpha(x) = \sum_{n,i,j \in \mathbb{Z}} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} c^n (-x_1)^{n-i} (-x_2)^{n-j} \frac{x^{\alpha+n+i+j+1}}{\alpha + n + i + j + 1}$$

with

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2c}.$$

*Proof sketch.*

1. Series expansion:

$$(1 + u)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} u^n \Rightarrow \text{convergence radius: } |u| < 1$$

2. Factorisation by roots:

$$ax + bx^2 + cx^3 = c(x - x_1)(x - x_2)x$$

3. Expansion of the roots:

$$(x - x_y)^n = \sum_{i=0}^n \binom{n}{i} x^i (-x_y)^{n-i}$$

□

## Radial Geodesics ( $L = 0$ )

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Energy conditions:

$$E^2 = \left( \frac{dr}{d\lambda} \right)^2 + \left( 1 - \frac{r_s}{r} \right)$$

Lorentzian length:

$$\lambda(p, s) = \frac{R^{\frac{3}{2}}}{2\sqrt{r_s}} (\pi - \eta(p) - \sin \eta(p))$$

$$\eta(x) = \arccos \left( \frac{2r_x}{R} - 1 \right)$$


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Time function:

$$E = \left( 1 - \frac{r_s}{r} \right) \frac{dt}{d\lambda}$$

$$t(p, s) = -E \sum_{n,m=0}^{\infty} \binom{-\frac{1}{2}}{n} \frac{r_s^{-n-m-\frac{3}{2}} k^n}{n+m+\frac{5}{2}} r_p^{n+m+\frac{5}{2}}$$

## Non-Radial Geodesics ( $L \neq 0$ )

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$$\lambda(p, s) = \frac{1}{L} \sum_{n,i,j} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} \left(\frac{1}{r_s}\right)^{n-\frac{1}{2}} \frac{(-r_1)^{-i} (-r_2)^{-j}}{n+i+j+\frac{5}{2}} r_p^{n+i+j+\frac{5}{2}}$$

$$\varphi(p, s) = \sum_{n,i,j} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} \left(\frac{1}{r_s}\right)^{n+\frac{1}{2}} \frac{(-r_1)^{-i} (-r_2)^{-j}}{n+i+j+\frac{1}{2}} r_p^{n+i+j+\frac{5}{2}}$$

$$t(p, s) = -\frac{E}{L} \sum_{m,n,i,j} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} \left(\frac{1}{r_s}\right)^{m+n+\frac{3}{2}} \frac{(-r_1)^{-i} (-r_2)^{-j}}{m+n+i+j+\frac{7}{2}} r_p^{m+n+i+j+\frac{7}{2}}$$

## Hypergeometric Function

$$F_1(a; b_1, b_2; c; x, y) = \sum_{m,n=0}^{\infty} \frac{(a)^{\overline{m+n}} (b_1)^{\overline{m}} (b_2)^{\overline{n}}}{(c)^{\overline{m+n}} m! n!} x^m y^n$$

$$G_1(\alpha; r) := \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \left(\frac{1}{r_s}\right)^{n+\frac{1}{2}} \frac{r^{\alpha+n}}{\alpha+n} F_1\left(\alpha+n; -n, -n; \alpha+n+1; \frac{r}{r_1}, \frac{r}{r_2}\right)$$

$$\lambda(p, s) = \frac{1}{L} G_1\left(\frac{5}{2}; r_p\right) \quad \Bigg| \quad \varphi(p, s) = G_1\left(\frac{1}{2}; r_p\right) \quad \Bigg| \quad t(p, s) = -\frac{E}{L} \sum_{m=0}^{\infty} r_s^{-m-1} G_1\left(m + \frac{7}{2}; r_p\right)$$

## Regularity at the Singular Boundary

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$$\xi : (r_p, E, L) \mapsto (r_p, t(r_p, E, L), \varphi(r_p, E, L))$$

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### Proposition

For  $r > 0$  satisfying

$$\left| -\frac{r}{r_s} + \frac{r^2}{L^2} + \frac{kr^3}{r_s L^2} \right| < 1,$$

and  $E \neq 0$  the Jacobian determinant is non zero, i.e.

$$\det(d\xi) \neq 0.$$

## Regularity at the Singular Boundary

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*Proof sketch.*

1. Differentiate explicit power series expressions.
2. Calculate  $J = \partial_E \varphi \partial_L t - \partial_L \varphi \partial_E t$
3. Start setting the first five coefficients to be zero and analyze the resulting condition.
4. Result:  $r_s = 0$  or  $r_1 = r_2 \Rightarrow \Delta = 0$ .

□



## Regularity at the Singular Boundary

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### Proposition

Differentiability of  $\lambda$  is ensured also for  $L \rightarrow 0$  and fixed  $r_p$ .

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*Proof sketch.* Rescale  $(r, r_p, L) \mapsto r_s(\varepsilon, \kappa, \delta)$ :

1. Differentiate the integral representation and express in relation to:

$$K_\alpha = r_s^{1-\alpha} \int_0^\kappa d\varepsilon \frac{\varepsilon^{\frac{9}{2}-\alpha}}{(k\varepsilon^3 + \varepsilon^2 - \delta^2\varepsilon + \delta^2)^{\frac{3}{2}}}, \quad \alpha \in \{0, 2, 3\}.$$

2. Anonymously evaluate  $0 \leq \varepsilon \leq \delta$  and  $\delta \leq \varepsilon \leq \kappa$  separately.

3. Use the notation

$$f(x, y) \underset{x}{\propto} x^a :\Leftrightarrow \forall y : f(x, y) \in \mathcal{O}(x^a),$$

to find

$$K_\alpha \underset{\delta}{\propto} \delta^{\min(\frac{5}{2}-\alpha, 0)}.$$

4. Use the chain rule with  $A = \begin{pmatrix} \partial_E \varphi & \partial_E t \\ \partial_L \varphi & \partial_L t \end{pmatrix}$ , and  $y^T = (\partial_r \varphi, \partial_r t)$ . Inverte it to find no divergence:

$$\begin{pmatrix} \partial_E \lambda \\ \partial_L \lambda \\ \partial_r \lambda \end{pmatrix} = \overbrace{\begin{pmatrix} A & 0 \\ v^T & 1 \end{pmatrix}}^{=d\xi} \begin{pmatrix} \partial_\varphi \lambda \\ \partial_t \lambda \\ \partial_r \lambda \end{pmatrix} \Rightarrow \begin{pmatrix} \partial_\varphi \lambda \\ \partial_t \lambda \\ \partial_r \lambda \end{pmatrix} = \begin{pmatrix} A^{-1} & 0 \\ -v^T A^{-1} & 1 \end{pmatrix} \begin{pmatrix} \partial_E \lambda \\ \partial_L \lambda \\ \partial_r \lambda \end{pmatrix} \underset{\delta}{\propto} \begin{pmatrix} \delta^1 \\ \delta^0 \\ \delta^0 \end{pmatrix}.$$

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### Proposition

Differentiability of  $\lambda$  is ensured also for  $(r_p, L) = r_s(\kappa, \delta) \mapsto (0, 0)$ .

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### Proposition

Differentiability of  $\lambda$  is ensured also for  $(r_p, L) = r_s(\kappa, \delta) \mapsto (0, 0)$ .

*Proof sketch.*

1. Consider the generalized integral

$$K_\alpha = r_s^{1-\alpha} \int_0^\kappa d\varepsilon \frac{\varepsilon^{\frac{9}{2}-\alpha}}{(k\varepsilon^3 + \varepsilon^2 - \delta^2\varepsilon + \delta^2)^{\frac{3}{2}}},$$

but this time separately evaluate  $\kappa < \delta$  and  $\delta < \kappa$ .

2. Same leading order  $\rightarrow$  introduce  $\chi := \max(\kappa, \delta)$  and find

$$K_{\alpha} \propto_{\chi} \chi^{\frac{5}{2}-\alpha}$$

by treating  $\sigma := \frac{\delta}{\kappa}$  and  $\chi$  as independent variables.

3. Explicitly determine  $\chi$ -proportionality of each differential.

4. Use the chain rule again, which takes the form:

$$\begin{pmatrix} \partial_{\varphi} \lambda \\ \partial_t \lambda \end{pmatrix} = A^{-1} \begin{pmatrix} \partial_E \lambda \\ \partial_L \lambda \end{pmatrix} = \frac{1}{K_0 K_3 + L^2 K_2 K_3} \begin{pmatrix} -E^2 L K_2 K_0 - L K_0 K_3 - L^3 K_2 K_3 \\ E K_0 K_3 + E L^2 K_2 K_3 \end{pmatrix} + \Xi,$$

where  $\Xi$  are higher order terms vanishing in the limit. The divergence cancels out.  $\square$

## Differentiability at the Singular Boundary

### Proposition

$\lambda$  is not differentiable with respect to  $r$ .

*Proof sketch.*

Let  $L > 0$  be fixed and  $r_p \rightarrow 0$ . Looking at the integral we know  $K_\alpha \propto_\kappa \kappa^{\frac{11}{2}-\alpha}$ .

Using the chain rule introduced earlier yields

$$\begin{pmatrix} \partial_\varphi \lambda \\ \partial_t \lambda \end{pmatrix} \propto_\kappa \kappa^{-6} \begin{pmatrix} \kappa^{-\frac{1}{2}} & \kappa^{\frac{5}{2}} \end{pmatrix} \begin{pmatrix} \kappa^{\frac{7}{2}} & \kappa^{\frac{7}{2}} \\ \kappa^{\frac{5}{2}} & \kappa^{\frac{7}{2}} \end{pmatrix} \begin{pmatrix} \kappa^{\frac{11}{2}} \\ \kappa^{\frac{5}{2}} \end{pmatrix} \propto_\kappa \begin{pmatrix} \kappa^{-\frac{1}{2}} \\ \kappa^{\frac{5}{2}} \end{pmatrix}.$$

□

## Differentiability at the Singular Boundary

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**Proposition**

Differentiability of  $\lambda$  is also assured with respect to  $\theta$ .



## Differentiability at the Singular Boundary

### Proposition

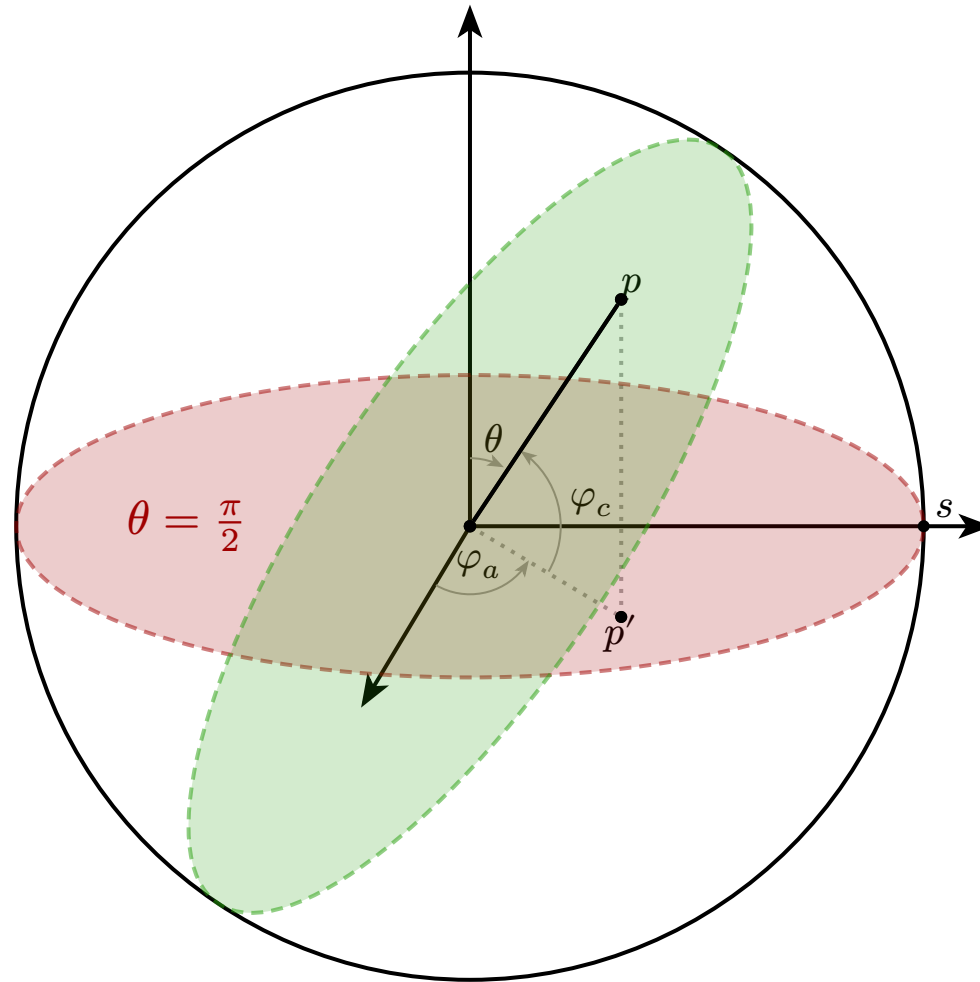
Differentiability of  $\lambda$  is also assured with respect to  $\theta$ .

*Proof sketch.* The angle so far denoted as  $\varphi$  is now  $\varphi_c$  and the actual one is  $\varphi_a$ . The relation between them is given by

$$\cos \varphi_c = \hat{e}_p \cdot \hat{e}_s = \cos \varphi_a \sin \theta.$$

By using the chain rule as before, every divergence cancels out.

The geometric idea:



## Conclusion

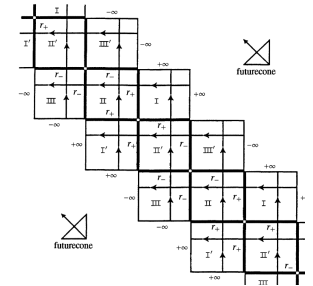
We found non-trivial regularity properties at the singular boundaries along  $\partial_\varphi$  and  $\partial_t$ .

**Next step:** Formulate a more generalized statement on the Differentiability of  $\tau$  with respect to  $\varphi$  and  $t$  for any neighbourhood  $U$  of  $q \in \mathcal{S}$ .

→ Investigate other solutions of the Einstein field equations (i.e. Kerr or Kerr-Newman<sup>3</sup>).

## Goal

Find a Synthetic extension to the Kruskal spacetime at  $r = 0$ .



<sup>3</sup>Barrett O'Neill, *The Geometry of Kerr Black Holes* / Barrett O'Neill (Wellesley, Mass., 1995).

<sup>4</sup>This presentation was created using the typst template "taupsi". Credit and thanks to Lyding Brumm for creating it.

*Proof.*

### Goal

$K_m$  admits no  $C^2$  isometric extension.

**Idea:** Follow  $C^0$ -inextendible future timelike geodesic curves!

Does the geodesic curve  $c : (a, b) \rightarrow K_m$  cross the horizon  $r = r_s$ ?

**Case 1** - yes:  $r \circ c(s) \xrightarrow{s \rightarrow b} 0$

**Case 2** - no:  $t \circ c(s) \rightarrow \infty \Rightarrow \ell(c) = \infty$

Kretschmann Scalar  $\mathcal{K} = R_{ijkl}R^{ijkl} = \frac{12r_s^2}{r^6} \xrightarrow{r \rightarrow 0} \infty$

□

*Proof.*

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□

**Definition** (Chronological Homeomorphism)

A chronological homeomorphism between two spacetimes  $M$  and  $N$  is a homeomorphism  $f : M \rightarrow N$  that preserves the chronological relation:

$$p \ll q \Leftrightarrow f(p) \ll f(q).$$

## Definition

The Appell series  $F_1$  is defined for  $|x| < 1$  and  $|y| < 1$  by

$$F_1(a; b_1, b_2; c; x, y) = \sum_{m, n=0}^{\infty} \frac{(a)^{\overline{m+n}} (b_1)^{\overline{m}} (b_2)^{\overline{n}}}{(c)^{\overline{m+n}} m! n!} x^m y^n.$$

## Derivatives

$$\frac{\partial^n}{\partial x^n} F_1(a, b_1, b_2, c; x, y) = \frac{(a)^{\overline{n}} (b_1)^{\overline{n}}}{(c)^{\overline{n}}} F_1(a + n, b_1 + n, b_2, c + n; x, y)$$

## Integral representation

$$F_1(a, b_1, b_2, c; x, y) = \frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)} \int_0^1 t^{a-1} (1-t)^{c-a-1} (1-xt)^{-b_1} (1-yt)^{-b_2} dt$$

with  $\Re c > \Re a > 0$



## Finiteness

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If we consider Radial geodesics starting from  $r_p = r_s$  at rest, we find:  $\lambda(p, s) = \frac{\pi r_s}{2}$ .

Metric:  $g = -\left(1 - \frac{r_s}{r}\right)dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1}dr^2 + r^2d\Omega$

For a future-directed causal curve  $\gamma : [r_p, r_q) \rightarrow M$  starting at a point  $p$  with  $\gamma(r_p) = p$ , one finds:

$$\lambda(p, q) = \int_{r_q}^{r_p} \sqrt{-g(\dot{\gamma}(r), \dot{\gamma}(r))} dr \leq \int_{r_q}^{r_p} \sqrt{\left(\frac{r_s}{r} - 1\right)^{-1}} dr.$$

Assume  $r < \frac{r_s}{2} \Rightarrow \lambda(p, q) < r_p - r_q \leq r_p$ .

**Continuity:**  $M_{\text{int}}$  is g.h., let  $s = (0, \varphi, \theta, t) \in \mathcal{S}$  and  $\varepsilon > 0$ .

- $U := I^+(\varepsilon, \varphi, \theta, t)$  open nbh. of  $s$
- $|\tau(p, x) - \tau(p, s)| \leq \varepsilon \forall x \in U$

*Proof sketch.* The Jacobi-determinant:  $J = \partial_E \varphi \partial_L t - \partial_L \varphi \partial_E t$

$$= \sum_{n,i,j,u,v,k,\ell} \binom{-\frac{1}{2}}{n} \binom{n}{i} \binom{n}{j} \binom{-\frac{1}{2}}{u} \binom{u}{k} \binom{u}{\ell} \frac{E}{L} \left( \frac{1}{r_s} \right)^{u+v+n+2} \frac{r_p^{n+i+j+u+v+k+\ell+4}}{(n + \frac{1}{2} + i + j)(u + \frac{7}{2} + v + k + \ell)} \\ \cdot \left[ (i\ell - jk) \left( \frac{-2E^2 r_s}{LE(E^2 - 1)\Delta} \right) - i \cdot \frac{r_s - 2E^2 r_s + 2\Delta}{2LE(E^2 - 1)\Delta} - j \cdot \frac{-r_s + 2E^2 r_s + 2\Delta}{2LE(E^2 - 1)\Delta} \right]$$

This is not well-defined for  $\Delta \rightarrow 0$ , but if we take a closer look at its origin:

$$\partial_L (r_1^i r_2^j) = \left[ (i - j) \frac{r_s}{2\Delta} + (i + j) \right] \frac{2}{L} k^{i+j-1} \cdot \left( \Delta - \frac{r_s}{2} \right)^{-i} \left( -\Delta - \frac{r_s}{2} \right)^{-j}$$

□

*Proof.* To get the actual  $\varphi_a$  from the formerly calculated  $\varphi_c$ , we must look at the projection of the point  $p$  into the  $(\theta = \frac{\pi}{2})$ -plane.

Consider

$$\cos \varphi_c = \hat{e}_p \cdot \hat{e}_s = \begin{pmatrix} \cos \varphi_a \sin \theta \\ \sin \varphi_a \sin \theta \\ \cos \theta \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \cos \varphi_a \sin \theta,$$

to find

$$\frac{\partial \varphi_a}{\partial \varphi_c} = \frac{\sin \varphi_c}{\sin \varphi_a \sin \theta}.$$

Now use the chain rule

$$\begin{pmatrix} \partial_\theta \lambda \\ \partial_E \lambda \\ \partial_L \lambda \\ \partial_r \lambda \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & \partial_\theta \varphi_a & 0 & 0 \\ 0 & \partial_E \varphi_a & \partial_E t & 0 \\ 0 & \partial_L \varphi_a & \partial_L t & 0 \\ 0 & \partial_r \varphi_a & \partial_r t & 1 \end{pmatrix}}_{=:B} \begin{pmatrix} \partial_\theta \lambda \\ \partial_{\varphi_a} \lambda \\ \partial_t \lambda \\ \partial_r \lambda \end{pmatrix},$$

and invert  $B$  with  $x^T := (\partial_\theta \varphi_a, 0)$  to find

$$B^{-1} = \begin{pmatrix} 1 & -x^T A^{-1} & 0 \\ 0 & A^{-1} & 0 \\ 0 & -v^T A^{-1} & 1 \end{pmatrix}.$$

Explicit calculation yields

$$A^{-1} = \frac{1}{J_c} \begin{pmatrix} \xi \partial_L t & -\xi \partial_E t \\ -\partial_L \varphi_c & \partial_E \varphi_c \end{pmatrix} \quad \text{with} \quad \xi := \frac{\sin \varphi_a \sin \theta}{\sin \varphi_c}$$

$$-x^T A^{-1} = \frac{\partial_L t, -\partial_E t}{J_c} \cdot \frac{\cos \varphi_a \cos \theta}{\sin \varphi_c},$$

where  $\varphi_c = 0 \Rightarrow \theta = \frac{\pi}{2} \Leftrightarrow \varphi_c = \varphi_a$

□